

ChadWallet — Build Notes

Live: chad-wallet-beryl.vercel.app · Repo: github.com/gauryvg98/chad.wallet · Engineering: chad-wallet-beryl.vercel.app/engineering

1 AI tooling & time

Primarily **Cursor with the Composer model**, and **Claude occasionally** for the harder reasoning — the cache/fan-out architecture, the swap-landing debugging, and the position cost-basis math. **~4 days** end-to-end.

Task	~Time
Scaffolding + landing page (fomo-style)	~5h
Trade page — token lists, chart, holders/trades, buy panel	~8h
Go backend — providers, cache-first pollers, WS hub	~8h
Non-custodial swap flow (build / sign / broadcast / confirm)	~4h
Portfolio + chain-derived positions & PnL	~4h
Real-time discovery (PumpPortal) + live prices / candles	~3h
Deploy (Vercel + Fly) + hardening	~2h
Polish — mobile, fomo parity, engineering doc	~4h

2 Issues & fixes

- **Free upstreams are rate-limited** (GeckoTerminal ~30/min). **Fix:** cache-first, poller-driven backend + one websocket hub that fans a single fetch out to every viewer — load scales with *tokens tracked*, not *users*.
- **Token-switching felt laggy**. **Fix:** optimistic active-token context (switches on click) + per-mint external store + debounced heavy fetches, so only the middle/side re-render, not the page.
- **Live prices caused re-render storms** across 50+ rows. **Fix:** useSyncExternalStore per mint + requestAnimationFrame batching, so only the rows that ticked repaint.
- **Charts looked sparse on 1m**. **Fix:** reconciled the lagging history feed against the live candle at the seam, and gave the currently-viewed chart priority on the upstream queue.
- **Swaps not landing first try**. **Fix:** dynamic slippage + priority-fee-to-land + rebroadcast-until-confirmed. Also diagnosed a 'sell won't land' as the wallet being out of SOL for the wrapped-SOL rent/fee — adding a sell-side gas guard.
- **Graduating tokens rendered blank**. **Fix:** PumpPortal migration events carry only the mint; backfill symbol/name/market-cap from pump.fun (the same enrichment path that resolves logos).
- **Mobile trade page felt like endless scroll**. **Fix:** buy panel moved under the price, feed height capped + internally scrollable, and the trending list became a slide-in drawer.

3 How it can be improved

- **Paid data tiers are the single biggest lever**. The whole thing runs on free/keyless sources, so upgrading (paid Helius, BirdEye / GeckoTerminal Pro, a dedicated RPC) **automatically improves the flow** — higher rate limits (no poller throttling), historical prices, full holder/PnL data, and complete swap history. Most of the items below fall out of it.
- **True mark-to-market equity curve** (per-token OHLCV × amount-held-over-time) — today's is a real but *stepped* realized-equity approximation; unlocked by historical-price access.
- **Real holder PnL & trader identities** via a swap indexer (Helius webhooks, indexed to a store) — currently clearly-labeled sample data.
- **Per-mint full swap history** endpoint — the activity feed is capped at recent swaps today.
- **Monetization:** on-chain per-trade fee via Jupiter platformFeeBps + a referral account.
- **Scale & durability:** Postgres (snapshots, accounts) + a Redis-backed cache/hub for horizontal scaling; Prometheus/Grafana observability.
- **Hardening:** exclude stablecoins/wSOL from position reconstruction, handle non-SOL-funded buy legs in cost basis, and ship the sell-side gas guard.